MetaCNVNet: A hybrid deep learning framework with multi-caller integration for enhanced copy number variation detection

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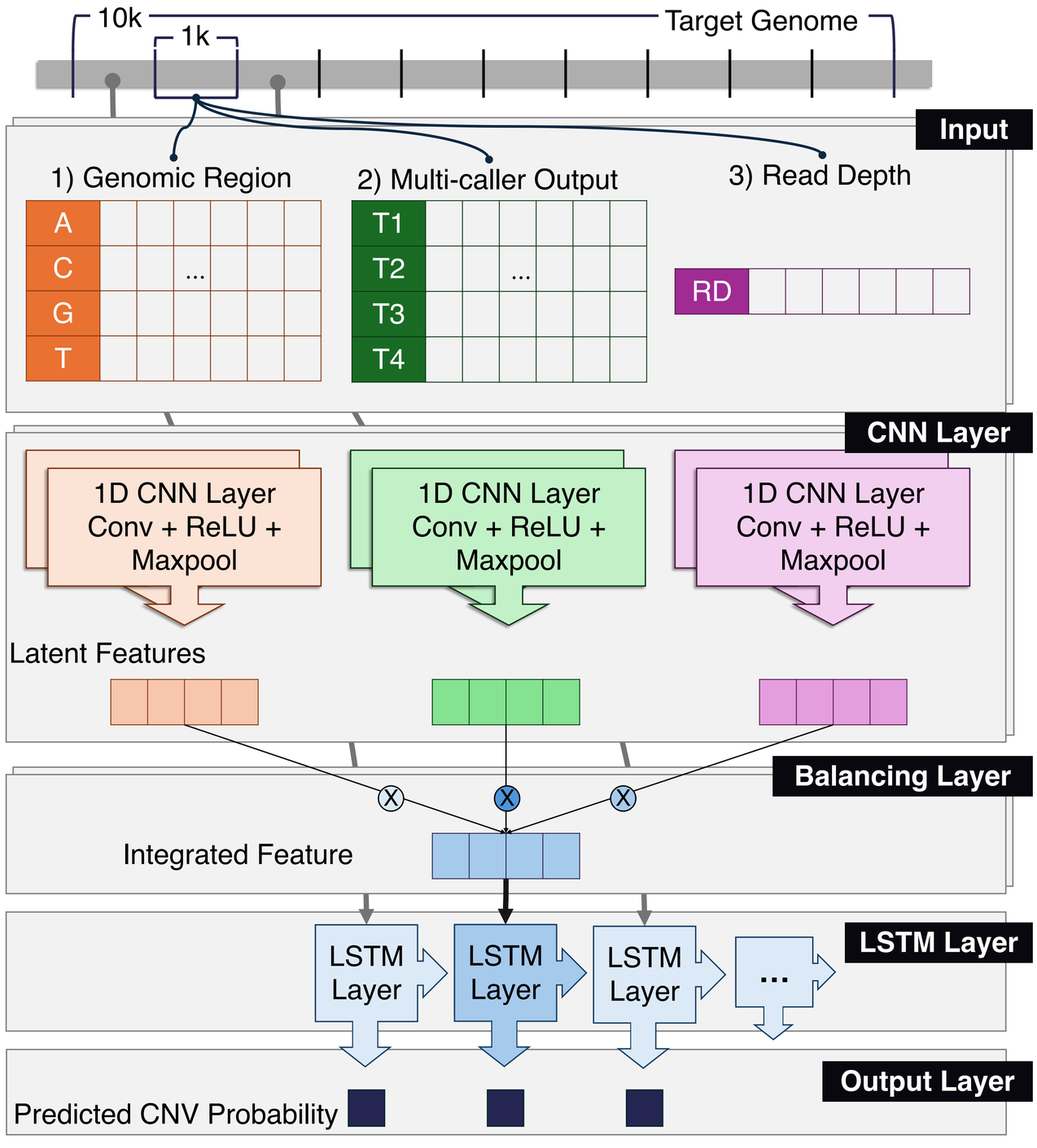
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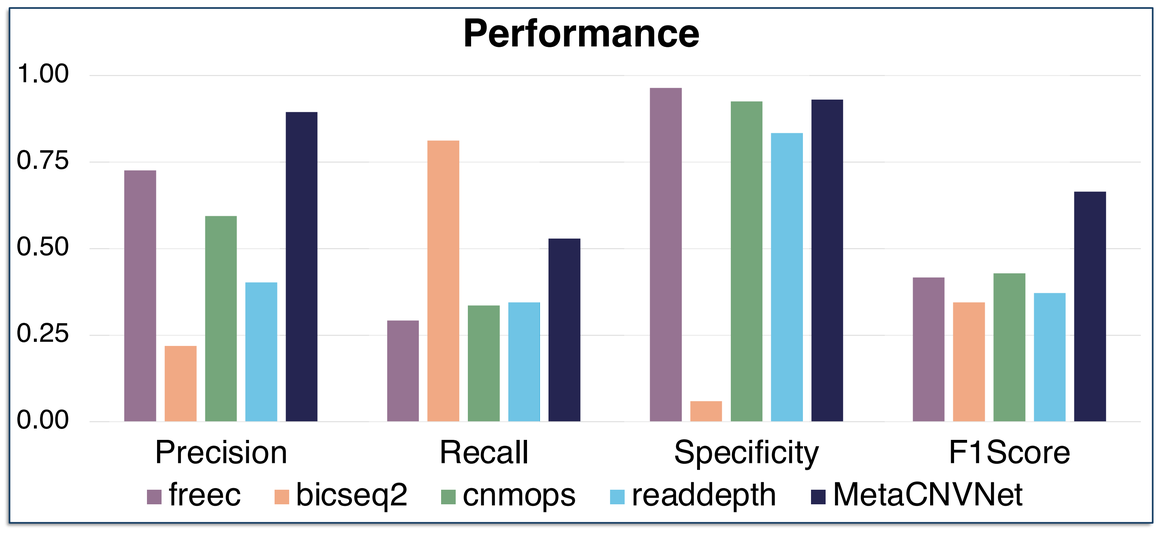
Background

- **Copy number variations (CNVs)** are structural DNA alterations ranging from kilobases to megabases, significantly contributing to genomic disorders and cancers through gene dosage effects and regulatory disruptions.
- Accurate detection of CNVs is essential for both clinical and research applications. However, current detection methods demonstrate inconsistent performance across different genomic contexts and sample qualities, with notable disparities in sensitivity and specificity despite numerous algorithmic advances.
- No gold standard method has been established for reliable CNV detection across diverse experimental settings, which continues to hinder consistent CNV identification and interpretation in genomic analysis.
- **MetaCNVNet** addresses these challenges through a hybrid deep learning approach that integrates multi-caller outputs with genomic sequence and read-depth features, substantially improving detection precision, especially for complex CNVs.

MetaCNVNet: CNN + LSTM Hybrid Model



- **Input**: Genomic sequences were one-hot encoded. Read depth (RD) signals computed from the read alignments, and multi-caller output aggregated from multiple CNV callers (READDEPTH[1], FREEC[2], CNVkit[3], CN.MOPS[4]).
- **CNN Layer**: Each input feature (sequence, RD, metadata profiles) was independently processed through two consecutive 1D convolutional layers (Conv → ReLU → MaxPool) to extract latent feature representations.
- **Balancing Layer**: A Balancing Layer learns feature-specific weights and integrates them into a single latent vector for each 1kb bin.
- **LSTM Layer**: The weighted feature vectors were sequentially fed into an LSTM network to capture contextual dependencies across multiple adjacent 1kb bins (e.g., 10kb in total), enabling prediction over longer genomic regions.
- **Output Layer**: Final CNV probability predictions (0–1) were generated through a dense layer followed by a sigmoid activation.



Conclusions

- **MetaCNVNet consistently outperforms CNV callers in precision, specificity, and F1-score by integrating multi-caller outputs with raw genomic features through a hybrid CNN–LSTM architecture.**
- By integrating diverse features, MetaCNVNet overcomes caller inconsistencies and lowers false positives
- By combining convolutional layers for local patterns and LSTM layers for long-range dependencies, MetaCNVNet robustly captures genomic signals for accurate CNV detection across diverse regions.
- At this stage, our model focuses on binary CNV detection at the 1k bin level. Future extensions will target finer-resolution modeling, long-range dependency capture, CNV subtype classification and copy number estimation to improve the model's broader applicability,
- Integration with long-read sequencing and domain adaptation techniques could broaden MetaCNVNet's applicability across diverse sample types and sequencing platforms.

References

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